Self-assessment questionnaire on programming strategies

Welcome to our study on the topic of “Self-assessment questionnaire on programming strategies”. We appreciate your interest and thank you for your participation. Below, we provide a brief overview of the study's content.

Background and objective of the study

Developing, understanding, or debugging software is a complex and highly individual process. In addition to applying specific programming techniques, programmers are constantly confronted with solving problems. Various personal strategies and approaches are used to do this. Psychology has researched and cataloged a variety of problem-solving strategies, from trial and error to goal-means analysis. However, there has been insufficient research to date on which of these basic psychological strategies programmers actually use in practice and to what extent these depend on the respective phase in the software development cycle. The aim of this study is to examine which specific strategies programmers use in different situations of the software development cycle and which of the underlying problem-solving strategies known in psychology can be assigned to these applied techniques.

Procedure

Once you have agreed to participate, you will be asked a few questions to help us assess your programming experience. You will then evaluate a series of statements that describe your approach to the programming process. Based on your answers, we will show you a derived profile of your problem-solving strategies (with percentages). Finally, you will evaluate the extent to which this calculated profile corresponds to your own assessment.

There are 37 questions in this survey.

programming experience

Please answer the following questions about your programming experience.

How many years have you been programming? *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ <1 year
- ◯ 1-5 years
- ◯ 5-10 years
- ◯ 10-15 years
- ◯ >15 years

How would you rate your own programming experience in relation to your fellow students or colleagues at work? *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ very inexperienced
- ◯ slightly less experienced
- ◯ similarly experienced
- ◯ slightly more experienced
- ◯ significantly more experienced

How often do you program? *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ once a month or less
- ◯ several times per month
- ◯ once a week
- ◯ several times per week
- ◯ daily

How many different projects have you been involved in as a programmer? (Including significant projects during your studies or in your private life) *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ less than 4
- ◯ between 4-7
- ◯ more than 7

preparation, planning, and design process

In the following section, you will evaluate your approach to preparation, planning, and the design process. For each statement, please assess whether it applies to you.

Please choose the appropriate response for each item:

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
I use agile principles (e.g., Scrum) to manage the overall process during development.	○	○	○	○	○	○
I use coordination rounds (e.g., daily meetings) to clarify the distribution of tasks.	○	○	○	○	○	○
I define the necessary tests for a function before I start with the actual code implementation.	○	○	○	○	○	○
I only implement new functionalities once all existing tests have been successfully completed.	○	○	○	○	○	○
I use well-known design patterns as proven solutions for recurring architectural problems.	○	○	○	○	○	○
I can precisely identify design patterns, find it in structured collections, and apply it.	○	○	○	○	○	○

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
I actively break down complex problems into smaller sub-problems that can be implemented independently of each other.	○	○	○	○	○	○
I begin a design task with a clear functional breakdown to define the overall structure.	○	○	○	○	○	○
I begin implementing a function by first determining what types it requires and returns before writing the function body.	○	○	○	○	○	○
I first define the required data types in order to clarify the logical structure of the problem top-down.	○	○	○	○	○	○
I deliberately define incomplete code sections (“program holes”) in order to first create the overall skeleton of the solution.	○	○	○	○	○	○
I implement the most difficult or complex parts of the code first.	○	○	○	○	○	○
I analytically weigh trade-offs between competing goals such	○	○	○	○	○	○

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
as performance and maintainability.						
My design decisions are based on a logical analysis of the advantages and disadvantages of each possible solution.	○	○	○	○	○	○

Do you have any thoughts, anecdotes, or other comments about the strategies mentioned? (e.g., under what circumstances you like to use them/don't like to use them)

Please write your answer here:

programming

In the following section, you will evaluate your approach to programming. For each statement, please assess whether it applies to you.

Please choose the appropriate response for each item:

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
I use detailed, step-by-step instructions to achieve consistent results.	○	○	○	○	○	○
I document my programs so that they can be clearly understood and executed by others.	○	○	○	○	○	○
I assign a starting value to all variables before using them in calculations.	○	○	○	○	○	○
When performing calculations, I actively look out for special cases that could lead to errors (e.g., division by zero).	○	○	○	○	○	○
I actively monitor my work progress and reflect on my goals in order to plan my next steps.	○	○	○	○	○	○
I organize my work steps and develop a work plan before I begin with the actual implementation.	○	○	○	○	○	○

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
I program larger blocks of code at once before executing the code.	○	○	○	○	○	○
When designing a program, I look for an existing code example in external sources to adapt it as a template.	○	○	○	○	○	○
I deliberately write experimental code to gain insights that I can later use to refine the final solution.	○	○	○	○	○	○
While programming, I write temporary notes directly into the code to structure my thoughts before I proceed with the final implementation.	○	○	○	○	○	○
I actively use LLM tools (e.g., ChatGPT) to generate suggested solutions for programming tasks.	○	○	○	○	○	○

Do you have any thoughts, anecdotes, or other comments about the strategies mentioned? (e.g., under what circumstances you like to use them/don't like to use them)

Please write your answer here:

program analysis, program comprehension

In the following section, you will evaluate your approach to program analysis and program comprehension. For each statement, please assess whether it applies to you.

Please choose the appropriate response for each item:

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
When navigating code, I let myself be guided by my subjective feeling as to which information is most likely to lead to the solution.	○	○	○	○	○	○
I try to recognize patterns and establish connections between different sections of code.	○	○	○	○	○	○
I use “divide and conquer methods” to understand the code.	○	○	○	○	○	○
I interpret compiler error messages as direct instructions for my next steps.	○	○	○	○	○	○
I use compiler errors to correct and refine my mental model of the code.	○	○	○	○	○	○
To understand code or find a starting point, I first interact with the program's user interface (UI).	○	○	○	○	○	○

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
I recognize familiar algorithmic patterns or design patterns and thus quickly grasp the higher semantic meaning of the code.	○	○	○	○	○	○
I can easily apply my abstract knowledge of code and design principles to new problems that are unfamiliar to me.	○	○	○	○	○	○
When I get stuck on a problem, I actively seek interaction with colleagues or online communities.	○	○	○	○	○	○
When encountering obstacles in program development, I actively seek assistance from teachers or experienced developers.	○	○	○	○	○	○

Do you have any thoughts, anecdotes, or other comments about the strategies mentioned? (e.g., under what circumstances you like to use them/don't like to use them)

Please write your answer here:

debugging

In the following section, you will evaluate your debugging approach. For each statement, please assess whether it applies to you.

Please choose the appropriate response for each item:

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
My debugging process is clearly divided into the phases of error localization, diagnosis of the cause, and correction.	○	○	○	○	○	○
When I am confronted with an error, I first formulate a hypothesis, test it, and then reject or modify it if it fails.	○	○	○	○	○	○
I use print statements or logging to observe the program flow and the state of the variables.	○	○	○	○	○	○
I use breakpoints and a debugger to examine the program state at specific points in time.	○	○	○	○	○	○
I trace the program execution step by step in order to systematically understand the control and data flow.	○	○	○	○	○	○
I visualize the memory (e.g., with a memory	○	○	○	○	○	○

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
table) to understand how variable values change during execution.						
When I encounter an error, I systematically work backwards from the incorrect output to the original source of the incorrect value.	○	○	○	○	○	○
I collect all possible causes of errors and then check them systematically.	○	○	○	○	○	○
I deliberately make random code changes and test immediately to see if the error disappears.	○	○	○	○	○	○
I use the program output to ask direct questions about the status of the program (e.g., “Why did this value arise?”).	○	○	○	○	○	○
I learn from mistakes so that I can avoid them in future projects.	○	○	○	○	○	○
I use techniques such as program slicing to reduce the amount of code to be examined to the parts relevant to the error.	○	○	○	○	○	○

	highly disagree	somewhat disagree	neutral	somewhat agree	highly agree	no answer
I use error localization tools that provide me with a ranking of the most probable error locations.	○	○	○	○	○	○
I rely on automated processes that flag code changes as suspicious in order to find the error more quickly.	○	○	○	○	○	○

Do you have any thoughts, anecdotes, or other comments about the strategies mentioned? (e.g., under what circumstances you like to use them/don't like to use them)

Please write your answer here:

Feedback

Based on your answers, the questionnaire calculates an assessment of how strongly certain types of solution strategies are pronounced in you.

Schema-driven resolution & domain-specific knowledge

This strategy means that you immediately recognize a type of problem and thereby activate a knowledge package (schema) that leads you directly to the solution method. You use your extensive experience to recognize patterns. You are likely to apply this strategy in areas where you have built up expertise.

NaN%

Please estimate how accurately this percentage allocation applies to you. *

Choose one of the following answers

Please choose **only one** of the following:

- ◯ too low
- ◯ fitting
- ◯ too high

Please provide brief feedback: How accurate do you think the assessment is? Does the description apply to your personal problem-solving style (for high percentages) or not (for low percentages)? Are you actually familiar with this way of thinking (for high percentages) or unfamiliar with it (for low percentages)?

Please write your answer here:

Feedback

Based on your answers, the questionnaire calculates an assessment of how strongly certain types of solution strategies are pronounced in you.

Problem decomposition & subgoals

You break down a large, complex problem into smaller, more easily solvable subproblems or steps. This significantly reduces the scope of the search for a solution. Often, this strategy also involves planning a suitable sequence in which these subgoals must be completed.

Please rate how applicable this problem-solving strategy is to you. *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ too low
- ◯ fitting
- ◯ too high

Please provide brief feedback: How accurate do you think the assessment is? Does the description apply to your personal problem-solving style (for high percentages) or not (for low percentages)? Are you actually familiar with this way of thinking (for high percentages) or unfamiliar with it (for low percentages)?

Please write your answer here:

Feedback

Based on your answers, the questionnaire calculates an assessment of how strongly certain types of solution strategies are pronounced in you.

Working Forward

You start with the given information or the initial state of the problem and apply operators (e.g., formulas) step by step, whose required variables are known. You work your way from the given situation to the goal.

Please rate how applicable this problem-solving strategy is to you. *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ too low
- ◯ fitting
- ◯ too high

Please provide brief feedback: How accurate do you think the assessment is? Does the description apply to your personal problem-solving style (for high percentages) or not (for low percentages)? Are you actually familiar with this way of thinking (for high percentages) or unfamiliar with it (for low percentages)?

Please write your answer here:

Feedback

Based on your answers, the questionnaire calculates an assessment of how strongly certain types of solution strategies are pronounced in you.

Working backwards

You start with the target state (or a sub-goal) and then look for the operations (operators) that could produce this state from a state closer to the beginning. In other words, you work backwards from the target to the initial state.

Please rate how applicable this problem-solving strategy is to you. *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ too low
- ◯ fitting
- ◯ too high

Please provide brief feedback: How accurate do you think the assessment is? Does the description apply to your personal problem-solving style (for high percentages) or not (for low percentages)? Are you actually familiar with this way of thinking (for high percentages) or unfamiliar with it (for low percentages)?

Please write your answer here:

Feedback

Based on your answers, the questionnaire calculates an assessment of how strongly certain types of solution strategies are pronounced in you.

Generate and test

First, you generate a series of possible solutions or hypotheses (e.g., causes of errors, scientific theories). Then you systematically test each of these generated solutions to find the correct one.

Please rate how applicable this problem-solving strategy is to you. *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ too low
- ◯ fitting
- ◯ too high

Please provide brief feedback: How accurate do you think the assessment is? Does the description apply to your personal problem-solving style (for high percentages) or not (for low percentages)? Are you actually familiar with this way of thinking (for high percentages) or unfamiliar with it (for low percentages)?

Please write your answer here:

Feedback

Based on your answers, the questionnaire calculates an assessment of how strongly certain types of solution strategies are pronounced in you.

Problem solving by analogy

You look for a similar problem that you have already successfully solved. Then you transfer the solution structure of this previous problem to the current problem. This strategy requires the ability to look beyond superficial similarities.

Please rate how applicable this problem-solving strategy is to you. *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ too low
- ◯ fitting
- ◯ too high

Please provide brief feedback: How accurate do you think the assessment is? Does the description apply to your personal problem-solving style (for high percentages) or not (for low percentages)? Are you actually familiar with this way of thinking (for high percentages) or unfamiliar with it (for low percentages)?

Please write your answer here:

Feedback

Based on your answers, the questionnaire calculates an assessment of how strongly certain types of solution strategies are pronounced in you.

Metacognitive and Affective Strategies

These strategies relate to how you manage and persevere with your own problem-solving process. They include self-regulation (e.g., monitoring your progress, assessing the difficulty of the task) and volitional elements (e.g., effort, persistence, belief in your own problem-solving ability).

Please rate how applicable this problem-solving strategy is to you. *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ too low
- ◯ fitting
- ◯ too high

Please provide brief feedback: How accurate do you think the assessment is? Does the description apply to your personal problem-solving style (for high percentages) or not (for low percentages)? Are you actually familiar with this way of thinking (for high percentages) or unfamiliar with it (for low percentages)?

Please write your answer here:

Feedback

Based on your answers, the questionnaire calculates an assessment of how strongly certain types of solution strategies are pronounced in you.

Planning strategy

You determine the goal and choose the right solution strategy. This serves as a guideline for the entire process and can, for example, involve breaking the problem down into sub-problems.

Please rate how applicable this problem-solving strategy is to you. *

Choose one of the following answers
Please choose **only one** of the following:

- ◯ too low
- ◯ fitting
- ◯ too high

Please provide brief feedback: How accurate do you think the assessment is? Does the description apply to your personal problem-solving style (for high percentages) or not (for low percentages)? Are you actually familiar with this way of thinking (for high percentages) or unfamiliar with it (for low percentages)?

Please write your answer here:

Feedback

Thank you for participating. Your answers provide us with valuable insights into the types of problem-solving strategies used in software development. If you have any questions about the survey or the research project, please feel free to contact us.

Florian Grabs

E-Mail: florian.grabs@s2018.tu-chemnitz.de

Belinda Schantong

E-Mail: belinda.schantong@informatik.tu-chemnitz.de

Janet Siegmund

E-Mail: janet.siegmund@informatik.tu-chemnitz.de

If you have any further comments or general feedback on the questionnaire, please feel free to enter them here:

Please write your answer here:

03-02-2026 – 16:17

Submit your survey.

Thank you for completing this survey.